A-level

# ENVIRONMENTAL SCIENCE

## Paper 2 - Generated practice paper Set 28

## Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance | Marks |
|---|---|---|
| **01.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Images / maps of large areas<br>– land-use change, deforestation, flooding<br>**1** Vegetation density and surface temperature<br>**1** Wind velocity, sea-surface altitude, wave height, oil pollution<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss | 3 |
| **01.2** | 1 mark for the point below.<br>**1** Microwaves pass through clouds (unlike visible light)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **01.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Very large / remote areas can be surveyed quickly and repeatedly so changes over time are detected<br>**1** No need for researchers to visit inaccessible areas, reducing cost and disturbance<br>– data are standardised across the whole area<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.<br>• 'trees give us the oxygen we breathe' as the main impact - credit carbon storage, soil protection, the water cycle and habitat loss | 2 |
| **01.4** | **Any 4 of 5**<br>1 mark each for any 4 of the 5 points below, in any order, maximum 4.<br>**1** Motion-sensitive cameras placed along trails record animals passing, using visible light by day and infrared at night<br>**1** Each tiger has a unique pattern of facial / body stripes<br>**1** Images are compared with the database so individuals are recognised rather than counted twice<br>**1** The number of different individuals recorded gives a minimum population<br>– repeated captures give territory size, movements, lifespans and social groupings<br>**1** **Limitation:** only works for species with unique visible features<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **infrared** → heat radiation / long-wave radiation re-emitted by the Earth | 4 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-403, 406-407; AQA spec 3.1.2.3.6. | |
| **02.1** | **Calculation**<br>A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark.<br>**1** 600 + (90 + 150) - (220 + 80) = 540<br>**1** Stock fell by 60 so the catch exceeds the MSY / unsustainable<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (thousand tonnes) is already printed on the answer line.<br>• **MSY** → the largest catch that can be taken without reducing the stock<br>**Do not credit**<br>• Points after the first 2 - the question asks for 2, so a list of extras cannot pick up the marks.<br>• A bare wrong answer with no working. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **02.2** | **Calculation**<br>1 mark for the correct answer; working does not have to be shown.<br>**1** A + G - M = 90 + 150 - 80 = 160<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>• The value alone - the unit (thousand tonnes) is already printed on the answer line.<br>**Do not credit**<br>• A bare wrong answer with no working. | 1 |
| **02.3** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Catch is landed and weighed at ports so it is recorded<br>**1** Other terms need data on fish in the sea: populations are mobile (crossing national boundaries), very unevenly distributed<br>**1** Representative sampling of a large enough proportion of the area is impractical / expensive<br>– survival of eggs and larvae varies greatly between years<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |
| **02.4** | **2 marks per point**<br>2 marks for each of the 2 points below; award 1 mark where a point is only partly made.<br>**2 CPUE:** if more effort (boat-days, engine power, net size) is needed to land the same catch, the population is declining even though total catch is maintained<br>– a better measure than catch size which technology can prop up<br>**2** Falling mean size / age (from scale or otolith growth rings) shows older adults are being removed faster than replaced and the catch depends on small young fish<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 4 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 349-351; AQA spec 3.5.2.2. | |
| **03.1** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Hydrosere<br>**1** Single-celled algae (and floating plants) - arrive on birds / insects / from soil<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **pioneer species** → the first species to colonise bare ground | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **03.2** | **1 mark each**<br>1 mark for each of the 5 points below.<br>**1** Pioneers and later plants grow and die<br>– dead organic matter plus sediment washed in fill the pond so water becomes shallower<br>**1** As depth falls rooted plants (pondweeds then emergent reeds with leaves above water) can anchor and colonise<br>– emergents shade submerged plants which die out<br>**1** Sediment / soil accumulates and becomes able to support larger plants<br>– water-tolerant trees (willow, alder) colonise the waterlogged soil<br>**1** Transpiration by trees and further sediment dry the soil and make it aerobic so oak can colonise<br>– the denser canopy shades out smaller plants<br>**1** Each community changes abiotic conditions (depth, light, nutrients, oxygen), making it suitable for the next colonisers which out-compete them<br>– conditions become less extreme and biotic factors increasingly important<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **waterlogged** → soil pores filled with water so there is no air<br>• **Transpiration** → water lost as vapour from the leaves<br>• **abiotic** → non-living / physical factors | 5 |
| **03.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Starting conditions (bare rock vs open water) differ but each stage makes conditions less extreme and builds soil / organic matter<br>**1** The final community is controlled by the regional climate (temperature, water availability), which is the same for both - climatic climax (eg oak woodland in the UK)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **climax community** → the stable final community<br>**Do not credit**<br>• 'the climax community never changes' - it is stable only while the conditions stay the same | 2 |
| **03.4** | 1 mark for the point below.<br>**1** Lichens and algae colonise bare rock (extreme temperatures, no soil, little water)<br>– rock fragments and dead organic matter accumulate<br>– mosses, then grasses and ferns as thin soil forms<br>– wind-dispersed trees (birch) then animal-dispersed (oak, beech)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 73-76; AQA spec 3.1.3.3. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **04.1** | **Correct answer only**<br>1 mark for the correct option. No mark if more than one box is ticked.<br>**1** B<br>**Also accept**<br>• **biodiversity** → the variety of life: habitat, species and genetic diversity<br>• **endemic** → found naturally only in that one place | 1 |
| **04.2** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1 Rainforest:** high light, warm, wet all year, stable conditions for long periods, continuous productivity, many niches (stratification)<br>**1 Antarctica:** very low temperatures, low precipitation, extreme seasonal light, 98% ice cover so almost no terrestrial life<br>**1** Antarctic biodiversity is marine, supported by upwelling nutrients and the summer bloom<br>– rainforest supports over half of terrestrial species on 10% of land<br>**1** Both are important carbon stores / climate regulators but rainforest species are specialists vulnerable to small changes<br>– Antarctic species adapted to extreme fluctuation<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• The converse of any point, ie the same comparison written the other way round.<br>• **biodiversity** → the variety of life: habitat, species and genetic diversity | 4 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4. | |
| **05.1** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** About 40-50 traps<br>**1** Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **reliable** → repeatable / gives consistent results when repeated<br>**Do not credit**<br>• 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument | 2 |
| **05.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Widely scattered results (large standard deviation) mean a single sample may not be representative<br>**1** More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **standard deviation** → a measure of the spread of the data about the mean | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **05.3** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1** Same type and size of pitfall trap, set for the same length of time<br>**1** Traps emptied at the same time of day / same weather conditions in both habitats<br>**1** Same number of traps and same method of positioning (random or systematic) in each habitat<br>**1** Other variables that could affect results (eg rainfall, temperature) monitored or controlled<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• The converse of any point, ie the same comparison written the other way round.<br>• **pitfall trap** → a sunken container that catches ground invertebrates<br>• **random** → positions chosen using random numbers so the choice is not biased | 3 |
| **05.4** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally<br>**1** Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends<br>**1** The interval between samples should match the rate at which the factor changes, found from a preliminary study<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1. | |
| **06.1** | 1 mark for the point below.<br>**1** Direct measurements of past conditions cannot be taken, so indirect evidence that is related to the condition must be used<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| **06.2** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** Age / date of the sample (radioisotope dating)<br>**1** Past temperatures<br>**1** Composition of the past atmosphere (eg $CO_2$ / methane concentrations)<br>**1** **Temperature:** more magnesium is incorporated into calcite at higher temperatures<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 4 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **06.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Movement of continents changed the position of land masses / latitude and therefore the climate and habitats present<br>**1** Ocean currents redistribute heat and nutrients<br>– changes in currents (eg as continents moved) altered climates, so past conditions cannot be explained without knowing how the systems worked<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **06.4** | **Any 3 of 4**<br>1 mark each for any 3 of the 4 points below, in any order, maximum 3.<br>**1** There was no oxygen / ozone layer so high levels of UV radiation reached the land surface<br>– water absorbed the UV and protected organisms in the oceans<br>**1** Warm temperatures around the vents and a rich mix of chemicals / increasingly complex organic molecules in the sea made biological reactions more likely<br>**1** The atmosphere contained toxic gases such as ammonia<br>– the early Archaea were anaerobic so did not need oxygen<br>**1** Water acted as the solvent for reactions / transport<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **ozone layer** → the ozone in the stratosphere that absorbs UV<br>**Do not credit**<br>• Points after the first 3 - the question asks for 3, so a list of extras cannot pick up the marks.<br>• answers that make ozone depletion a cause of global warming, or that describe CFCs as physically blowing a hole - chlorine atoms catalyse the breakdown of ozone and are regenerated | 3 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 6-9; AQA spec 3.1.1.1-3.1.1.3. | |
| **07.1** | **Calculation**<br>A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark.<br>**1 Subsistence maize:** 9.0 / 0.6 = 15<br>**1 Intensive wheat:** 54.0 / 18.0 = 3<br>**1 Feedlot beef:** 12.0 / 40.0 = 0.3<br>**Also accept**<br>• The correct answer however it is set out, including a different but valid method.<br>• An answer carried forward from an earlier incorrect step (error carried forward).<br>• An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify.<br>**Do not credit**<br>• A bare wrong answer with no working. | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **07.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** An energy subsidy is the energy put into a farming system by people rather than by the Sun: fuel for machinery, the manufacture of fertilisers and pesticides, irrigation, transport, drying and storage<br>**1** Intensive systems raise the yield per hectare by adding large energy subsidies, so the output rises but the input rises faster and the ratio falls<br>**1** The feedlot ratio is below 1, so it takes more energy to produce the beef than the beef contains<br>– the extra trophic level means most of the energy in the feed is lost in respiration, excretion and the inedible parts of the animal<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **trophic level** → the feeding level in a food chain<br>**Do not credit**<br>• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on<br>• 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts | 3 |
| **07.3** | **Any 3 of 6**<br>1 mark each for any 3 of the 6 points below, in any order, maximum 3.<br>**1** Keep the food chain short by producing crops for direct human consumption rather than feeding them to livestock, so one transfer of about 10% efficiency is avoided<br>**1** Restrict the movement of livestock, so less of the energy in the feed is used for muscular activity and more is converted to biomass<br>**1** Keep the animals warm, or rear them indoors, so less energy is spent maintaining body temperature<br>**1** Slaughter the animals young, while their growth rate and food conversion ratio are still high<br>**1** Use species or breeds with a high food conversion ratio, such as poultry or farmed fish, rather than cattle<br>**1** Control parasites and disease, and use feed that is easy to digest, so more of the food eaten is assimilated<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **food conversion ratio** → the feed needed per unit of food produced<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.<br>• 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts | 3 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **07.4** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** A single desired gene can be transferred, so the characteristic appears without the unwanted genes that come with a whole parent in conventional breeding<br>**1** Genes can be moved between species that could never interbreed, so characteristics can be introduced that selective breeding could never produce, and the result is achieved in one generation instead of many<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| **07.5** | **Any 4 of 6**<br>1 mark each for any 4 of the 6 points below, in any order, maximum 4.<br>**1** Genes may escape by pollen transfer to wild relatives or to neighbouring non-GM crops, creating herbicide-resistant weeds or making organic certification impossible<br>**1** Insect-resistant crops may kill non-target species, including the pollinators and natural pest predators that the wider ecosystem depends on<br>**1** Pests and weeds evolve resistance to the toxin or the herbicide the crop is designed to be used with, so the advantage is temporary and use of the chemical increases<br>**1** Reliance on a few patented GM varieties narrows the genetic diversity of the crop, making it more vulnerable to a new disease<br>**1** Seed is patented and must be bought each year, which can put subsistence farmers in debt or make them dependent on one supplier<br>**1** Long-term effects on human health are not fully known, and labelling may not let consumers choose<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **genetic diversity** → the range of alleles / genes present in a population<br>**Do not credit**<br>• Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.<br>• 'GM crops are unnatural' as an environmental impact - credit gene flow, resistance, effects on non-target species or reduced pesticide use | 4 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **08.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Irrigation depletes rivers and aquifers (eg Aral Sea, Ogallala)<br>– increases evaporation<br>**1** Soil erosion and compaction reduce water retention and infiltration so runoff and river flow fluctuations increase<br>**1 Evapotranspiration changes:** increased in irrigated arid areas, decreased where forests were cleared<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **infiltration** → water soaking into the soil surface<br>• **Evapotranspiration** → evaporation plus transpiration<br>**Do not credit**<br>• 'the soil loses its nutrients' as a description of erosion - erosion is the physical loss of the soil itself | 3 |
| **08.2** | **2 marks per point**<br>2 marks for each of the 2 points below; award 1 mark where a point is only partly made.<br>**2 Seasonal / local:** avoids energy for heating, lighting greenhouses out of season and long-distance transport (food miles) and the pollution this causes<br>**2 Organic:** uses natural processes (rotation, legumes, manure, biological control, predator habitats) instead of artificial fertilisers and pesticides, so less pollution, energy use and harm to non-target species<br>– free-range and Fairtrade also accepted for welfare / social reasons<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **biological control** → using a natural predator, parasite or pathogen of the pest<br>**Do not credit**<br>• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on | 4 |
| **08.3** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Food shortages after 1945 (dependence on US food aid)<br>– farmers could not afford loans to invest so grants funded machinery, drainage, hedgerow removal, liming, improved livestock<br>**1 Guaranteed market / price support:** government bought surpluses to hold prices up (and sold stores in poor years) giving farmers security to increase output<br>**1 Consequences:** loss of hedgerows and wetlands, intensification, pesticide and fertiliser pollution<br>– later food surpluses (butter mountains) that could not be sold without harming LEDC producers<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on | 3 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-333; AQA spec 3.5.1.3-3.5.1.4. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **09.1** | **Any 2 of 6**<br>1 mark each for any 2 of the 6 points below, in any order, maximum 2.<br>**1** Hydrological cycle<br>**1** Carbon cycle<br>**1** Nitrogen cycle<br>**1** Atmospheric circulation<br>**1** Thermohaline circulation<br>**1** Photosynthesis<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **Thermohaline** → the ocean conveyor driven by temperature and salinity<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.<br>• 'plants breathe in carbon dioxide and breathe out oxygen' - name photosynthesis and respiration | 2 |
| **09.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Enzymes reduce the activation energy of reactions<br>**1** So reactions such as photosynthesis, decomposition and nitrogen fixation do not need high temperatures<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds | 2 |
| **09.3** | **Any 1 of 3**<br>1 mark each for any 1 of the 3 points below, in any order, maximum 1.<br>**1** Haber process (nitrogen fixation)<br>**1** Incineration of wastes (decomposition)<br>**1** Many manufacturing processes<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds<br>• **Incineration** → burning waste, usually with energy recovery | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **10.1** | **Levels of response**<br>Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist.<br>**Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.<br>**Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.<br>**Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.<br>**0 marks**: No relevant content.<br>**Indicative content - a guide to placing the mark, not a checklist**<br>• **Resources:** wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food<br>– new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions<br>• **Biomimetics:** vehicle design, architecture, structures, adhesion, materials, ultrasound<br>• Medicines from plants and animals (examples)<br>– most species unstudied - argument for habitat conservation<br>• Physiological research using wildlife species<br>– wildlife as pest control agents (ladybirds, Encarsia, Cactoblastis, Bt)<br>• **Genetic resources:** crop wild relatives, Vavilov centres, gene pools<br>• **Ecosystem services:** atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision<br>• Ethical / aesthetic / recreational arguments<br>– the unknown value of species not yet investigated (precautionary principle)<br>• **Evaluation:** which arguments are most persuasive to decision makers<br>– economic valuation vs intrinsic value<br>– judgement<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• Any accurate, relevant material that is not in the indicative content.<br>• **biodiversity** → the variety of life: habitat, species and genetic diversity<br>**Do not credit**<br>• the precautionary principle described as banning anything new - it shifts the burden of proof onto showing that harm is unlikely | 9 |
| | **Model answer - Level 3 (8 marks)** | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | Human society depends on other species first as resources: wood, fibres such as cotton, wool and silk, oils, fuels and, above all, food, with wild species such as the morama bean, perennial cereals, eland and bison offering new crops and livestock better adapted to local conditions. Biomimetics copies adaptations into vehicle design, architecture, structures, adhesives, materials and ultrasound; medicines such as taxol, quinine and AZT came from plants and animals, and physiological research on marsupials, hippos and sponges continues to yield medical insights. Wild species are pest-control agents - ladybirds, *Encarsia*, *Cactoblastis*, *Bacillus thuringiensis* - and crop wild relatives in Vavilov centres hold the genes future crops will need. | |
| | Less visibly, other species provide the services that keep the planet habitable: regulating atmospheric composition, driving the hydrological and biogeochemical cycles, maintaining soil, pollinating crops, dispersing seeds and providing habitat. Most species have never been investigated, so their value is unknown, and there are ethical, aesthetic and recreational reasons for conservation that owe nothing to usefulness. | |
| | Which arguments persuade decision-makers? Economic valuation of resources and services speaks the language of policy and has been most effective, but it undervalues the services that go unnoticed because they operate in dynamic equilibrium and it can never capture the value of species not yet studied. The strongest case combines the precautionary logic of unknown value with the intrinsic argument: conserving biodiversity protects both what we know we need and what we cannot yet know. | |
| | **Weaker answer - Level 1 (3 marks)** | |
| | Biodiversity is important for humans in many ways. We get food, wood and cotton from plants and animals. Many medicines come from plants. Bees pollinate our crops so we need to conserve them. Animals and plants are also nice to look at and people like to visit nature. We should conserve biodiversity because we might need species in the future. | |
| | **What makes the difference** | |
| | • Level 1 names categories without examples; Level 3 gives specific, accurate examples in every category. | |
| | • Level 3 includes ecosystem services, genetic resources and pest control that Level 1 omits. | |
| | • 'Discuss' at the top level means weighing the arguments against each other - Level 3 evaluates which arguments persuade and why; Level 1 does not. | |
| **10.2** | 1 mark for the point below.<br>**1** The variety and abundance of life (species, genes, habitats) in an area / on Earth<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **biodiversity** → the variety of life: habitat, species and genetic diversity | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| 11.1 | **Discuss how ecological monitoring and the development of new technologies have improved the planning and evaluation of conservation strategies.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• **Why monitor:** species present, population size, density, distribution, survival, age structure, breeding rates, movements<br>– baseline and evaluation<br>– Red List categorisation and quotas rely on it<br>• **Standard techniques and their limitations:** quadrats, transects, nets, traps, Lincoln index, Simpson's index, DAFOR<br>– sampling design (random / systematic, size, number, timing, standardisation)<br>• **New technologies:** satellite / radio / GPS / acoustic tracking, geolocators, camera traps, CCTV, image recognition databases, DNA databases, eDNA, sonograms, drones / ROVs / AUVs, satellite sensors, data loggers<br>• **How they improve validity:** representative data over large areas and times, less disturbance, individual identification, detection of hidden species, objectivity<br>• **Examples:** penguin colonies from satellites, tiger camera traps, ivory DNA, MCZ monitoring, released animals tracked | 25 |
| | • **Limitations:** cost, battery / tag weight, data handling, ethics, still need ground truthing<br>• Judgement<br>*Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*<br>**Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-407; AQA spec 3.1.2.3.5-3.1.2.3.6, 3.7. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| 11.2 | **Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• Dynamic equilibria and negative feedback in climate, hydrological cycle and populations<br>– positive feedback and tipping points caused by humans<br>• **Energy:** solar-driven, low energy density, low temperature (photosynthesis, enzymes, nitrogen fixation vs Haber process)<br>– renewable energy, heat pumps, bioleaching, fuel cells<br>• **Materials:** cycles, few abundant elements, biodegradable non-toxic wastes vs fossil fuels and dispersed minerals<br>– recycling, Cradle to Cradle, industrial symbiosis<br>• **Diversity and stability:** rainforests / reefs vs agroecosystems<br>– diverse energy mixes<br>• Optimum vs maximum production<br>– supporting ecosystems in agriculture, forestry, fisheries, urban design<br>• Case studies and metrics<br>– limits and judgement<br>*Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* | 25 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-379; AQA spec 3.6.1-3.6.4. | |

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